United States Patent Application Publication	20250285699
Kind Code	A1
Publication Date	September 11, 2025
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TEST DEVICE AND TEST METHOD FOR MEASURING TIMING SPECIFICATIONS OF MEMORY DEVICE

Abstract

A test device for a memory device includes a first exclusive OR (XOR) gate configured to output a first operation signal by performing an XOR operation between a clock signal and an output signal output from the memory device, an oscillator configured to output an oscillation signal, and an output circuit configured to determine a period of the oscillation signal. The output circuit may be configured to count a number of rising edges of the oscillation signal while the first operation signal is maintained at a high level, and determine an access time of the memory device based on the counted number of the rising edges, and a period of the oscillation signal.

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Appl. No.: **18/970531**

Filed: **December 05, 2024**

Foreign Application Priority Data

KR	10-2024-0033402	Mar. 08, 2024

Publication Classification

Int. Cl.: **G11C29/10** (20060101); **G01R31/3177** (20060101)

U.S. Cl.:

CPC **G11C29/10** (20130101); **G01R31/3177** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based on and claims priority under 35 USC § 119 to Korean Patent Application No. 10-2024-0033402, filed on Mar. 8, 2024, in the Korean Intellectual Property Office, the disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND

[0002] Example embodiments relate to a test device and a test method for measuring timing specifications of a memory device.

[0003] Semiconductor memory devices are classified into volatile memory devices, such as static random access memory (SRAM) devices or dynamic random access memory (DRAM) devices, and nonvolatile memory devices, such as flash memory devices, phase-change random access memory (PRAM) devices, magnetic random access memory (MRAM) devices, resistive random access memory (RRAM) devices, or ferroelectric random access memory (FRAM) devices. Volatile memory devices lose their stored data when their power supplies are interrupted, while nonvolatile memory devices retain their stored data even when their power supplies are interrupted.

[0004] In memory devices, defects may occur due to various factors. Various defects in memory devices, such as write defects in memory cells, may occur probabilistically and thus memory devices need to be tested.

[0005] For example, there is a growing demand for test devices and test methods for accurately measuring timing specifications including a read speed, an access time, a setup time, or a hold time of memory devices to ensure normal operations of the memory devices.

SUMMARY

[0006] Example embodiments provide a test device for accurately measuring timing specifications of a memory device.

[0007] According to one or more example embodiments, a test device for a memory device includes a first exclusive OR (XOR) gate configured to output a first operation signal by performing an XOR operation between a clock signal and an output signal output from the memory device, an oscillator configured to output an oscillation signal, and an output circuit configured to determine a period of the oscillation signal. The output circuit may be configured to count a number of rising edges of the oscillation signal while the first operation signal is maintained at a high level, and determine an access time of the memory device based on the counted number of the rising edges, and a period of the oscillation signal.

[0008] According to one or more example embodiments, a test method for a memory device includes determining a period of an oscillation signal, output from an oscillator; counting a number of rising edges of the oscillation signal that occur while a first operation signal, which is a result of an exclusive OR (XOR) operation between a clock signal and an output signal output from the memory device, is maintained at a high level, and determining an access time of the memory device based on the counted number of the rising edges, and the period of the oscillation signal.

[0009] According to one or more example embodiments, a test device for a memory device includes an oscillator configured to output an oscillation signal, a first exclusive OR (XOR) gate configured to perform an XOR operation between a clock signal and an output signal output from the memory device, a first chain circuit and a second chain circuit configured to output delayed clocks in which different delay times are respectively applied to the clock signal, a second XOR gate configured to perform an XOR operation between a signal output from the first chain circuit and a signal output from the second chain circuit, and an output circuit configured to determine a period of the oscillation signal. The output circuit may be configured to determine an access time of the memory device based on a number of rising edges of the oscillation signal that occur while a first operation signal output from the first XOR gate is maintained at a high level, and the period of the oscillation signal, and determine a unit delay time based on a number of rising edges of the oscillation signal that occur while a second operation signal output from the second XOR gate is maintained at a high level.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] The above and other objects and features of the present disclosure will become apparent by describing in detail example embodiments thereof with reference to the accompanying drawings.

[0011] FIG. **1** is a block diagram of a memory system according to one or more example embodiments.

[0012] FIG. **2**A is a circuit diagram of a memory system including a test device for measuring an access time of a memory device according to one or more example embodiments.

[0013] FIG. **2**B is a diagram illustrating a configuration in which the test device of FIG. **2**A determines an access time of the memory device.

[0014] FIG. **3**A is a circuit diagram of a memory system including a test device, the test device including a first chain circuit and a second chain circuit, according to one or more example embodiments.

[0015] FIG. **3**B is a diagram illustrating a configuration in which a control logic circuit according to one or more example embodiments controls a first chain circuit and a second circuit to output delay clocks to which different delay times are applied.

[0016] FIG. **3**C is a diagram illustrating a configuration in which the test device of FIG. **3**A determines a unit delay time of the first chain circuit and the second chain circuit.

[0017] FIG. **4**A is a circuit diagram illustrating a configuration in which a test device outputs a third delay clock and a fourth delay clock, to which different delay times are applied, through a first chain circuit.

[0018] FIG. **4**B is a diagram illustrating a configuration in which a first threshold setup time is determined through an output circuit when the test device of FIG. **4**A outputs a third delay clock through the first chain circuit.

[0019] FIG. **4**C is a diagram illustrating a configuration in which a second threshold setup time is determined through an output circuit when the test device of FIG. **4**A outputs a fourth delay clock through the first chain circuit.

[0020] FIG. **5**A is a circuit diagram illustrating a configuration in which a test device according to one or more example embodiments outputs a fifth delay clock and a sixth delay clock, to which different delay times are applied, through a first chain circuit.

[0021] FIG. **5**B is a diagram illustrating a configuration in which first threshold hold time is determined through an output circuit when the test device of FIG. **5**A outputs a third delay clock through the first chain circuit.

[0022] FIG. **5**C is a diagram illustrating a configuration in which second threshold hold time is determined through the output circuit when the test device of FIG. **5**A outputs a fourth delay clock through the first chain circuit.

[0023] FIG. **6**A is a circuit diagram illustrating a configuration of the first chain circuit according to one or more example embodiments.

[0024] FIG. **6**B is a circuit diagram illustrating a configuration of a first internal multiplexer and the second internal multiplexer of the first chain circuit of FIG. **6**A.

[0025] FIG. **7** is a circuit diagram illustrating a memory system including a test device further

including a second multiplexer according to one or more example embodiments.
[0026] FIG. **8** is a circuit diagram illustrating a memory system including a test device further including a third multiplexer according to one or more example embodiments.
[0027] FIG. **9** is a flowchart illustrating a test method for determining an access time of a memory device according to one or more example embodiments.
[0028] FIG. **10** is a flowchart illustrating a method for determining a unit delay time of a first chain circuit and a second chain circuit included in a test device according to one or more example embodiments.
[0029] FIG. **11** is a flowchart illustrating a test method for determining a setup time of a memory device based on unit delay time according to one or more example embodiments.
[0030] FIG. **12** is a flowchart illustrating a test method for determining a hold time of a memory device based on the unit delay time according to one or more example embodiments.
DETAILED DESCRIPTION
[0031] Hereinafter, example embodiments will be described with reference to the accompanying drawings.
[0032] FIG. **1** is a block diagram illustrating a memory system according to one or more example embodiments.
[0033] Referring to FIG. **1**, a memory system **100** according to one or more example embodiments may include a test device **110** and a memory device **120**.
[0034] The memory system **100** may include a memory device **120** configured to store input data.
[0035] For example, the memory device **120** may store data input from an outside. In addition, the memory device **120** may output stored data in response to a write request input from the outside.
[0036] The memory device **120** may include a nonvolatile memory such as, for example but not limited to, a flash memory, a magnetic RAM (MRAM), a ferroelectric RAM (FeRAM), a phase-change RAM (PRAM), or a resistive RAM (ReRAM), but example embodiments are not limited thereto. Examples of the memory device **120** may include, for example but not limited to, a dynamic random access memory (DRAM) such as a double data rate synchronous dynamic random access memory (DDR SDRAM), a low power double data rate (LPDDR) SDRAM, a graphics double data rate (GDDR) SDRAM, or a Rambus dynamic random access memory (RDRAM).
[0037] In addition, the memory system **100** may include a test device **110** electrically connected to the memory device **120**.
[0038] For example, the test device **110** may transmit a clock signal CK and an input signal Din to the memory device **120**. In addition, the test device **110** may obtain an output signal Dout output from the memory device **120**.
[0039] The test device **110** may perform a test operation on the memory device **120**.
[0040] For example, the test device **110** may determine (or measure) timing specifications of the memory device **120** using at least a portion of the clock signal CK, the input signal Din, and the output signal Dout.
[0041] According to one or more example embodiments, the test device **110** may determine an access time of the memory device **120** based on a result of an exclusive OR (XOR) operation between the clock signal CK and the output signal Dout.
[0042] For example, the test device **110** may determine the access time of the memory device **120** by performing a logical conjunction (AND) operation on the result of an XOR operation between the clock signal CK and the output signal Dout and an oscillation signal output from an oscillator.
[0043] For example, the access time may be understood as corresponding to a time taken for the memory device **120** to output an output signal Dout in response to a clock signal CK from the memory device **120** in a state in which the input signal Din is input to the memory device **120**.
[0044] For example, the access time may be understood as corresponding to a time from when the clock signal CK is input to the memory device **120** to when the output signal Dout is output from the memory device **120**, in a state in which the input signal Din is input to the memory device **120**.

[0045] For example, the access time may be understood as corresponding to a time taken for the memory device **120** to output data in response to an input command or store data in response to a write command.
[0046] According to one or more example embodiments, the test device **110** may determine (or measure) at least a portion of a setup time and a hold time of the memory device **120** using the clock signal CK and input signal Din.
[0047] For example, the test device **110** may determine a unit delay time based on a result of an XOR operation between delay clocks generated by applying different delay times to the clock signal CK.
[0048] For example, the test device **110** may determine a unit delay time by performing a logical conjunction (AND) operation on a result of an XOR operation between the delay clocks, generated by applying different delay times to the clock signal CK, and the oscillation signal output from the oscillator.
[0049] Furthermore, the test device **110** may determine at least a portion of the setup time and the hold time of the memory device **120** based on the clock signal CK, the input signal Din, and the unit delay time.
[0050] For example, the test device **110** may determine the setup time and/or the hold time of the memory device **120** based on the XOR operation result between the clock signal CK with a fixed delay time and the input signal Din, and the unit delay time.
[0051] For example, the setup time of the memory device **120** may refer to a minimum time during which a logic level of the input signal Din should be maintained before a rising or falling edge of the clock signal CK applied to the memory device **120**.
[0052] For example, the setup time of the memory device **120** may refer to a minimum amount of time required for the input signal Din to be determined as having a logic low level or a logic high level.
[0053] The hold time of a memory device **120** may refer to a minimum amount of time during which a logic level of the input signal Din should be maintained after the rising or falling edge of the clock signal CK applied to the memory device **120**.
[0054] For example, the hold time of the memory device **120** may refer to a minimum time required for the input signal Din to be output.
[0055] Based on the above-described configurations, the test device **110** may measure the timing specifications of the memory device **120** based on at least a portion of the input signal Din and clock signal CK input to the memory device **120** and the output signal Dout output from the memory device **120**.
[0056] Therefore, the test device **110** may also be referred to as, for example, a built-in self-test (BIST) device, a design for test (DFT) device, or the like.
[0057] In addition, referring to the above-described configurations, the test device **110** may measure the timing specifications of the memory device **120** based on the oscillation signal output from the oscillator.
[0058] As a result, the memory system **100** (or the test device **110**) according to one or more example embodiments may accurately measure the timing specifications of the memory device **120**.
[0059] FIG. **2**A is a circuit diagram of a memory system including a test device for measuring a access time of a memory device according to one or more example embodiments, and FIG. **2**B is a diagram illustrating a configuration in which the test device of FIG. **2**A determines the access time of the memory device.
[0060] Referring to FIGS. **2**A and **2**B, a memory system **100**A according to one or more example embodiments may include a test device **110**A and a memory device **120**.
[0061] The memory system **100**A illustrated in FIG. **2**A may be understood as an example of the memory system **100** illustrated in FIG. **1**.

[0062] Therefore, the same or substantially the same components may be represented by the same reference numerals, and redundant descriptions are omitted to avoid repetition.
[0063] Referring to FIG. **2**A, the test device **110**A according to one or more example embodiments may include an oscillator **220**, a first XOR gate XOR1, and an output circuit **230**. In addition, the test device **110**A may further include an AND gate A electrically connected between the first XOR gate XOR1 and the oscillator **220**, and the output circuit **230**.
[0064] The test device **110**A may include the first XOR gate XOR1 configured to perform an exclusive OR (XOR) operation between the clock signal CK and the output signal Dout output from the memory device **120**.
[0065] The output signal Dout may be understood as a data signal output by the memory device **120** in response to the input signal Din and the clock signal CK.
[0066] For example, the first XOR gate XOR1 may perform an XOR operation between the clock signal CK and the output signal Dout to output a first operation signal CS**1**.
[0067] Referring to FIG. **2**B, for example, the first operation signal CS**1** may have a waveform maintained at a high level (for example, "1") during a period AT, which is from a time when a rising edge of the clock signal CK occurs to a time when a rising edge of the output signal Dout occurs.
[0068] In addition, the test device **110**A may include the oscillator **220** configured to output an oscillation signal OCS.
[0069] The oscillator **220** according to one or more example embodiments may output an oscillation signal OCS having a specified period. For example, the oscillator **220** may output an oscillation signal OCS having rising edges that occur according to the specified period.
[0070] For example, the oscillator **220** may include a plurality of inverters connected in series. The oscillator **220** may be referred to as a ring oscillator. However, the type and configuration of the oscillator **220** are not limited to the above-described example, and the oscillator **220** may be understood as having various types and configurations that may generate an oscillation signal OCS having a specific period.
[0071] In addition, the test device **110**A may include the output circuit **230** configured to determine a period of the oscillation signal OCS.
[0072] For example, the output circuit **230** may divide the oscillation signal OCS to determine the period of the oscillation signal OCS.
[0073] For example, the output circuit **230** may divide the oscillation signal OCS to determine that the period of the oscillation signal OCS is 100 picoseconds (ps). For example, the output circuit **230** may be referred to as a divider.
[0074] In addition, the output circuit **230** may count a number of rising edges of a first output signal OS**1** during the period AT.
[0075] For example, the test device **110**A may include an AND gate A configured to perform a logical conjunction (AND) operation between the first operation signal CS**1** and the oscillation signal OCS.
[0076] For example, the AND gate A may output the first output signal OS**1**, which is a result of the AND operation between the first operation signal CS**1** and the oscillation signal OCS.
[0077] Referring to FIG. **2**B, the first output signal OS**1** may have the same waveform as the waveform of the oscillation signal OCS while the first operation signal CS**1** is maintained at a high level. While the output circuit **230** may divide the oscillation signal OCS to determine the period of the oscillation signal OCS, for example, 100 ps, division of the oscillation signal OCS is omitted for brevity of description.
[0078] For example, the output circuit **230** may count a number of rising edges of the oscillation signal OCS (or the first output signal OS**1**) that occur while the first operation signal CS**1** is maintained at a high level.
[0079] For example, the output circuit **230** may be referred as a counter configured to count a

number of rising edges of an input signal thereof.
[0080] Furthermore, the output circuit **230** may determine the access time of the memory device **120** based on the number of rising edges of the oscillation signal OCS counted while the first operation signal CS**1** is maintained at a high level.
[0081] For example, the output circuit **230** may determine an access time of the memory device **120** based on the number of rising edges of the oscillation signal OCS, counted while the first operation signal CS**1** is maintained at a high level, and the period of the oscillation signal OCS.
[0082] For example, referring to FIG. **2**B, the output circuit **230** may determine that five rising edges of the oscillation signal OCS occurs while the first operation signal CS**1** is maintained at a high level.
[0083] Furthermore, the output circuit **230** may determine the access time of the memory device **120** to be 500 ps, based on five rising edges of the oscillation signal OCS that occur while the first operation signal CS**1** is maintained at a high level and the period of the oscillation signal OCS being 100 ps.
[0084] For example, the output circuit **230** may determine that the time taken for the memory device **120** to output an output signal Dout in response to the rising edge of the clock signal CK in a state, in which the input signal Din is input to the memory device **120**, is 500 ps.
[0085] Referring to the above-described configurations, the test device **110**A according to one or more example embodiments may divide the oscillation signal OCS to determine the period of the oscillation signal OCS.
[0086] In addition, the test device **110**A may measure the access time of the memory device **120** using the determined period and the result of the XOR operation between the clock signal CK and the output signal Dout.
[0087] As a result, the memory system **100**A according to one or more example embodiments may improve the accuracy of the operation or process of measuring the access time of the memory device **120**.
[0088] FIG. **3**A is a circuit diagram of a memory system **100**B including a test device including a first chain circuit and a second chain circuit according to one or more example embodiments. FIG. **3**B is a diagram illustrating a configuration in which a control logic circuit according to one or more example embodiments controls a first chain circuit and a second circuit to output delay clocks to which different delay times are applied. FIG. **3**C is a diagram illustrating a configuration in which the test device of FIG. **3**A determines a unit delay time of the first chain circuit and the second chain circuit. FIG. **4**A is a circuit diagram illustrating a configuration in which a test device outputs a third delay clock and a fourth delay clock, to which different delay times are applied, through a first chain circuit. FIG. **4**B is a diagram illustrating a configuration in which a first threshold setup time is determined through an output circuit when the test device of FIG. **4**A outputs a third delay clock through the first chain circuit. FIG. **4**C is a diagram illustrating a configuration in which a second threshold setup time is determined through an output circuit when the test device of FIG. **4**A outputs a fourth delay clock through the first chain circuit.
[0089] Referring to FIGS. **3**A to **3**C and FIGS. **4**A to **4**C, the memory system **100**B according to one or more example embodiments may include a test device **110**B and a memory device **120**.
[0090] The test device **110**B may include a first chain circuit **211**, a second chain circuit **212**, a control logic circuit **210**, a first multiplexer MUX**1**, a second XOR gate XOR2, an oscillator **220**, an AND gate A, and an output circuit **230**.
[0091] The test device **110**B according to one or more example embodiments may further include a clock generator **240** configured to generate a clock signal CK. The clock generator **240** may generate a clock signal CK having a specified frequency using a phase locked loop (PLL) circuit and/or an oscillator circuit.
[0092] The memory system **100**B illustrated in FIGS. **3**A and **4**A may be understood as an example of the memory system **100** illustrated in FIG. **1**.

[0093] Therefore, the same or substantially the same components may be represented by the same reference numerals, and redundant descriptions are omitted to avoid repetition.
[0094] According to one or more example embodiments, the test device **110**B may include the first chain circuit **211** and the second chain circuit **212**, each being configured to the clock signal CK.
[0095] For example, the test device **110**B may include the first chain circuit **211** configured to receive the clock signal CK and including a plurality of inverters. In addition, the test device **110**B may include a second chain circuit **212** configured to receive the clock signal CK and including a plurality of inverters.
[0096] For example, the first chain circuit **211** and the second chain circuit **212** may be implemented with substantially the same configuration. For example, at least a portion of the plurality of inverters included in each of the first chain circuit **211** and the second chain circuit **212** may be connected in series.
[0097] In addition, the test device **110**B may include the control logic circuit **210** electrically connected to the first chain circuit **211** and the second chain circuit **212**.
[0098] For example, the control logic circuit **210** may control at least a portion of the first chain circuit **211** and the second chain circuit **212** to output delay clocks DS**1** and DS**2** obtained by applying different delay times to the clock signal CK.
[0099] According to one or more example embodiments, the control logic circuit **210** may control the first chain circuit **211** to output a first delay clock DS**1** to which a first delay time, equal to a first integer multiple of a unit delay time, is applied.
[0100] For example, the control logic circuit **210** may control the first chain circuit **211** to output the clock signal CK, input to the first chain circuit **211**, through an electrical path including at least a portion of the plurality of inverters included in the first chain circuit **211**.
[0101] The unit delay time may be understood as corresponding to a delay time caused by one of the plurality of inverters.
[0102] Referring to FIG. **3**B, the control logic circuit **210** may control a first chain multiplexer CM**1** to output, as the first delay clock DS**1**, the clock signal CK, input to a first chain circuit **211**A, through a 1-4-th path P**14** including eight inverters connected in series.
[0103] For example, the control logic circuit **210** may control the first chain circuit **211**A to output, through the first chain multiplexer CM**1**, the first delay clock DS**1**, in which a first delay time equal to 8 times the unit delay time is applied to the clock signal CK.
[0104] In addition, the control logic circuit **210** may control the second chain circuit **212** to output a second delay clock DS**2** to which a second delay time, equal to a second integer multiple of the unit delay time, is applied.
[0105] For example, the control logic circuit **210** may control the second chain circuit **212** to output the clock signal CK, input to the second chain circuit **212**, through an electrical path including at least a portion of the plurality of inverters included in the second chain circuit **212**.
[0106] Referring to FIG. **3**B, the control logic circuit **210** may control a second chain multiplexer CM**2** to output, as the second delay clock DS**2**, the clock signal CK, input to a second chain circuit **212**A, through a 2-1-th path P**21** including two inverters connected in series.
[0107] For example, the control logic circuit **210** may control the second chain circuit **212**A to output, through the second chain multiplexer CM**2**, the second delay clock DS**2**, in which a delay time, equal to 2 times the unit delay time, is applied to the clock signal CK.
[0108] In addition, the test device **110**B may include the first multiplexer MUX**1** electrically connected between the second chain circuit **212** and the memory device **120**.
[0109] For example, the first multiplexer MUX**1** may selectively output one of the input signal Din and the second delay clock DS**2**.
[0110] For example, the first multiplexer MUX**1** may output the second delay clock DS**2** to the memory device **120** in response to a first select signal SEL**1**.
[0111] In addition, the test device **110**B may include the second XOR gate XOR2 configured to

perform an XOR operation between a signal, output from the first chain circuit **211**, and a signal output from the second chain circuit **212**.
[0112] For example, the test device **110**B may include the second XOR gate XOR2 configured to perform an XOR operation between a signal, output from the first chain circuit **211**, and a signal output from the first multiplexer MUX**1**.
[0113] According to one or more example embodiments, the second XOR gate XOR2 may perform an XOR operation between the first delay clock DS**1** and the second delay clock DS**2**.
[0114] For example, the second XOR gate XOR2 may output a second operation signal CS**2** as a result of an XORing the first delay clock DS**1** and the second delay clock DS**2**.
[0115] For example, referring to FIG. **3**C, the second XOR gate XOR2 may output a second operation signal CS**2** having a waveform maintained at a high level (for example, "1") in a period TD, which is from a time a rising edge of the second delay clock DS**2** occurs to a time a rising edge of the first delay clock DS**1** occurs.
[0116] In addition, the output circuit **230** may count a number of rising edges of a second output signal OS**2** during the period TD.
[0117] For example, the test device **110**B may include an AND gate A configured to perform an AND operation between the second operation signal CS**2** and the oscillation signal OCS. For example, the AND gate A may output the second output signal OS**2**, based on a result of the AND operation between the second operation signal CS**2** and the oscillation signal OCS.
[0118] Referring to FIG. **3**C, the second output signal OS**2** may have the same waveform as the waveform of the oscillation signal OCS while the second operation signal CS**2** is maintained at a high level.
[0119] The output circuit **230** may count the number of rising edges of the oscillation signal OCS that occur while the second operation signal CS**2** is maintained at a high level.
[0120] Furthermore, the output circuit **230** may determine a unit delay time based on the number of rising edges of the oscillation signal OCS that are counted while the second operation signal CS**2** is maintained at a high level.
[0121] For example, the output circuit **230** may determine a unit delay time corresponding to a delay time caused by a single inverter, based on the number of rising edges of the oscillation signal OCS that are counted while the second operation signal CS**2** is maintained at a high level and a period of the oscillation signal OCS.
[0122] For example, referring to FIG. **3**C, the output circuit **230** may determine that 12 rising edges of the oscillation signal OCS occur while the second operation signal CS**2** is maintained at a high level.
[0123] In addition, the output circuit **230** may determine that a time difference TD between the first delay clock DS**1** and the second delay clock DS**2** is 1200 ps, based on 12 rising edges of the oscillation signal OCS that occur while the second operation signal CS**2** is maintained at a high level and a period of the oscillation signal OCS being 100 ps.
[0124] Furthermore, the output circuit **230** may determine a unit delay time based on a difference between a number of inverters, included in an electrical path through which the first delay clock DS**1** is output, and a number of inverters included in an electrical path through which the second delay clock DS**2** is output.
[0125] For example, the first delay clock DS**1** may be output through the 1-4-th path P**14** including eight inverters connected in series, and the second delay clock DS**2** may be output through the 2-1-th path P**21** including two inverters connected in series. When the first delay clock DS**1** and the second delay clock DS**2** have a delay time difference corresponding to six inverters therebetween, the output circuit **230** may determine that the unit delay time is 200 ps by dividing 1200 ps, the time difference TD of the two delay clocks DS**1** and DS**2**, by 6.
[0126] For example, the output circuit **230** may determine that a unit delay time caused by a single inverter, among a plurality of inverters included in each of the first chain circuit **211** and the second

chain circuit **212**, is 200 ps.

[0127] Referring to the above-described configurations, the test device **110**B according to one or more example embodiments may determine the unit delay time using the period the oscillation signal OCS and a result of an XOR operation between and the delay clocks DS**1** and DS**2** generated by applying different delay times to the clock signal CK.

[0128] The unit delay time may be understood as a time corresponding to a delay time caused by a single inverter included in each of the first chain circuit **211** and the second chain circuit **212**.

[0129] Referring to FIG. **4**A, the control logic circuit **210** may control the first multiplexer MUX**1** to output an input signal Din to the memory device **120**.

[0130] In addition, the control logic circuit **210** according to one or more example embodiments may control the first chain circuit **211** to output a third delay clock DS**3** to which a third delay time, equal to a third integer multiple of the unit delay time, is applied.

[0131] For example, referring to FIG. **4**B, the control logic circuit **210** may control the first chain circuit **211** to output the third delay clock DS**3** generated by advancing a clock signal CK by a first time T**1**, which is equal to twice the unit delay time.

[0132] According to one or more example embodiments, the second XOR gate XOR2 may perform an XOR operation between the third delay clock DS**3** and an input signal Din. For example, the second XOR gate XOR2 may output a third operation signal CS**3** as a result of an XORing the third delay clock DS**3** and the input signal Din.

[0133] For example, the second XOR gate XOR2 may perform an XOR operation between the third delay clock DS**3** and the input signal Din based on the memory device **120** outputting an output signal Dout having a high level in response to the third delay clock DS**3**.

[0134] Referring to FIG. **4**B, the second XOR gate XOR2 may output a third operation signal CS**3** having a waveform maintained at a high level (for example, “1”) from a time a rising edge of the input signal Din occurs to a time a rising edge of the third delay clock DS**3** occurs.

[0135] In addition, the output circuit **230** may count a number of rising edges of the third output signal OS**3**.

[0136] The AND gate A may output the third output signal OS**3**, based on a result of an AND operation between the third operation signal CS**3** and the oscillation signal OCS.

[0137] Referring to FIG. **4**B, the third output signal OS**3** may have the same waveform as the waveform of the oscillation signal OCS while the third operation signal CS**3** is maintained at a high level.

[0138] For example, the output circuit **230** may count a number of rising edges of the oscillation signal OCS (or the third output signal OS**3**) that occur while the third operation signal CS**3** is maintained at a high level.

[0139] Furthermore, the output circuit **230** may determine a first threshold setup time ST**1** based on the number of rising edges of the oscillation signal OCS, which are counted while the third operation signal CS**3** is maintained at a high level, and a period of the oscillation signal OCS.

[0140] For example, referring to FIG. **4**B, the output circuit **230** may determine that four rising edges of the oscillation signal OCS occur while the third operation signal CS**3** is maintained at a high level.

[0141] In addition, the output circuit **230** may determine that a first threshold setup time ST**1** is 400 ps, based on four rising edges of the oscillation signal OCS that occur while the third operation signal CS**3** is maintained at a high level and the period of the oscillation signal OCS being 100 ps.

[0142] In addition, the control logic circuit **210** may control the first chain circuit **211** to output a fourth delay clock DS**4**, to which a fourth delay time equal to a fourth integer multiple of the unit delay time is applied, based on the memory device **120** outputting an output signal Dout having a high level in response to the third delay clock DS**3**.

[0143] For example, referring to FIG. **4**C, the control logic circuit **210** may control the first chain circuit **211** to output the fourth delay clock DS**4** generated by advancing the clock signal CK by a

second time T**2**, which is equal to four times the unit delay time.
[0144] Furthermore, the second XOR gate XOR2 may perform an XOR operation between the fourth delay clock DS**4** and the input signal Din.
[0145] For example, the second XOR gate XOR2 may output a fourth operation signal CS**4** as a result of an XORing the fourth delay clock DS**4** and the input signal Din.
[0146] For example, referring to FIG. **4**C, the second XOR gate XOR2 may output the fourth operation signal CS**4** having a waveform maintained at a high level (for example, “1”) from a time the rising edge of the input signal Din occurs to a time a rising edge of the fourth delay clock DS**4** occurs.
[0147] In addition, the output circuit **230** may count a number of rising edges of the fourth output signal OS**4**.
[0148] The AND gate A may output the fourth output signal OS**4**, based on a result of an AND operation between the fourth operation signal CS**4** and the oscillation signal OCS.
[0149] Referring FIG. **4**C, the fourth output signal OS**4** may have the same waveform as the waveform of the oscillation signal OCS while the fourth operation signal CS**4** is maintained at a high level.
[0150] For example, the output circuit **230** may count a number of rising edges of the oscillation signal OCS (or the fourth output signal OS**4**) that occur while the fourth operation signal CS**4** is maintained at a high level.
[0151] Furthermore, the output circuit **230** may determine a second threshold setup time ST**2** based on the number of rising edges of the oscillation signal OCS that are counted while the fourth operation signal CS**4** is maintained at a high level, and the period of the oscillation signal OCS.
[0152] For example, referring to FIG. **4**C, the output circuit **230** may determine that three rising edges of the oscillation signal OCS occur while the fourth operation signal CS**4** is maintained at a high level.
[0153] In addition, the output circuit **230** may determine that a second threshold setup time ST**2** is 300 ps, based on four rising edges of the oscillation signal OCS occur while the fourth operation signal CS**4** is maintained at a high level, and the period of the oscillation signal OCS being 100 ps.
[0154] Referring to FIG. **4**B, the memory device **120** may output an output signal Dout having a high-level (for example, “1”) in response to the input signal Din and the third delay clock DS**3**. The input signal Din and the third delay clock DS**3** may be understood to be in a state satisfying the setup time of the memory device **120**.
[0155] Referring to FIG. **4**C, the memory device **120** may output an output signal Dout having a low level (for example, “0”) in response to the input signal Din and the fourth delay clock DS**4**. The input signal Din and the fourth delay clock DS**4** may be understood to be in a setup time violation state that does not satisfy the setup time of the memory device **120**.
[0156] Accordingly, the test device **110**B may determine that the setup time of the memory device **120** has a value between the first threshold setup time ST**1** and the second threshold setup time ST**2**.
[0157] For example, the test device **110**B (or the control logic circuit **210**) may determine that the setup time of the memory device **120** has a value between a time difference between the rising edge of the third delay clock DS**3** and the rising edge of the input signal Din (corresponding to the first threshold setup time ST**1**) and a time difference between the rising edge of the fourth delay clock DS**4** and the rising edge of the input signal Din (corresponding to the second threshold setup time ST**2**).
[0158] Referring to the above-described configurations, the test device **110**B may determine a unit delay time, caused by an inverter included in each of the chain circuit **211** and **212**, using a result of an XOR operation between the delay clocks DS**1** and DS**2** to which different delay times are applied.
[0159] Furthermore, the test device **110**B may determine the setup time of the memory device **120**

based on the result of the XOR operation between the input signal Din and the delay clocks DS**3** and DS**4**, and the unit delay time.
[0160] In an example embodiment, the test device **110**B (or the control logic circuit **210**) may apply different delay times to the input signal Din to generate a plurality of delayed input signals.
[0161] For example, the test device **110**B may apply different delay times to the input signal Din such that the rising edge of the input signal Din is delayed by different delay times.
[0162] Furthermore, the test device **110**B may determine the setup time of the memory device **120** based on a result of an XOR operation between each of a plurality of delayed input signals and the clock signal CK, and the unit delay time.
[0163] As a result, the memory system **100**B according to one or more example embodiments may improve an accuracy of an operation (or process) of measuring the setup time of the memory device **120**.
[0164] FIG. **5**A is a circuit diagram illustrating a configuration in which a test device according to one or more example embodiments outputs a fifth delay clock and a sixth delay clock, to which different delay times are applied, through a first chain circuit. FIG. **5**B is a diagram illustrating a configuration in which first threshold hold time is determined through an output circuit when the test device of FIG. **5**A outputs a third delay clock through the first chain circuit. FIG. **5**C is a diagram illustrating a configuration in which second threshold hold time is determined through the output circuit when the test device of FIG. **5**A outputs a fourth delay clock through the first chain circuit.
[0165] Referring to FIGS. **5**A to **5**C, a memory system **100**B according to one or more example embodiments may include a test device **110**B and a memory device **120**.
[0166] It will be understood that the memory system **100**B illustrated in FIG. **5**A may have substantially the same configuration as the memory system **100**B illustrated in FIGS. **3**A and **4**A.
[0167] Therefore, the same or substantially the same components may be represented by the same reference numerals, and redundant descriptions are omitted to avoid repetition.
[0168] Referring to FIG. **5**A, the control logic circuit **210** may control the first multiplexer MUX**1** to output the input signal Din to the memory device **120**.
[0169] In addition, the control logic circuit **210** according to one or more example embodiments may control the first chain circuit **211** to output a fifth delay clock DS**5** to which a fifth delay time, equal to a fifth integer multiple of the unit delay time, is applied.
[0170] For example, referring to FIG. **5**B, the control logic circuit **210** may control the first chain circuit **211** to output a fifth delay clock DS**5** obtained by delaying the clock signal CK by a first time T**1** equal to twice the unit delay time.
[0171] According to one or more example embodiments, a second XOR gate XOR2 may perform an XOR operation between a fifth delay clock DS**5** and an input signal Din. For example, a second XOR gate XOR2 may output a fifth operation signal CS**5** as a result of XORing the fifth delay clock DS**5** and the input signal Din.
[0172] For example, the second XOR gate XOR2 may perform an XOR operation between the fifth delay clock DS**5** and the input signal Din based on the memory device **120** outputting an output signal Dout having a high level in response to the fifth delay clock DS**5**.
[0173] Referring to FIG. **5**B, the second XOR gate XOR2 may output a fifth operation signal CS**5** having a waveform maintained at a high level (for example, “1”) from a time a rising edge of the fifth delay clock DS**5** occurs to a time a falling edge of the input signal Din occurs.
[0174] In addition, the output circuit **230** may count a number of rising edges of a fifth output signal OS**5**.
[0175] An AND gate A may output the fifth output signal OS**5**, based on a result of an AND operation between the fifth operation signal CS**5** and the oscillation signal OCS.
[0176] Referring to FIG. **5**B, the fifth output signal OS**5** may have the same waveform as the waveform of the oscillation signal OCS while the fifth operation signal CS**5** is maintained at a high

level.
[0177] For example, the output circuit **230** may count a number of rising edges of the oscillation signal OCS (or the fifth output signal OS**5**) that occur while the fifth operation signal CS**5** is maintained at a high level.
[0178] Furthermore, the output circuit **230** may determine a first threshold hold time HT**1** based on the number of rising edges of the oscillation signal OCS that are counted while the fifth operation signal CS**5** is maintained at a high level, and a period of the oscillation signal OCS.
[0179] For example, referring to FIG. **5**B, the output circuit **230** may determine that three rising edge of the oscillation signal OCS occurs while the fifth operation signal CS**5** is maintained at a high level.
[0180] In addition, the output circuit **230** may determine that the first threshold hold time HT**1** is 300 ps based on three rising edges of the oscillation signal OCS that occur while the fifth operation signal CS**5** is maintained at a high level, and the period of the oscillation signal OCS being 100 ps.
[0181] In addition, the control logic circuit **210** may control the first chain circuit **211** to output a sixth delay clock DS**6**, to which a sixth delay time equal to a sixth integer multiple of the unit delay time is applied, based on the memory device **120** outputting an output signal Dout having a high level in response to the fifth delay clock DS**5**.
[0182] For example, referring to FIG. **5**C, the control logic circuit **210** may control the first chain circuit **211** to output a sixth delay clock DS**6** obtained by delaying the clock signal CK by a second time T**2** equal to four times the unit delay time.
[0183] Furthermore, the second XOR gate XOR2 may perform an XOR operation between the sixth delay clock DS**6** and the input signal Din.
[0184] For example, the second XOR gate XOR2 may output a sixth operation signal CS**6** as a result of XORing the sixth delay clock DS**6** and the input signal Din.
[0185] For example, referring to FIG. **5**C, the second XOR gate XOR2 may output a sixth operation signal CS**6** having a waveform maintained at a high level (for example, “1”) from a time a rising edge of the sixth delay clock DS**6** occurs to a time a falling edge of the input signal Din occurs.
[0186] In addition, the output circuit **230** may count a number of rising edges of a sixth output signal OS**6**.
[0187] The AND gate A may output the sixth output signal OS**6**, a result of an AND operation between the sixth operation signal CS**6** and the oscillation signal OCS.
[0188] Referring to FIG. **5**C, the sixth output signal OS**6** may have the same waveform as the waveform of the oscillation signal OCS while the sixth operation signal CS**6** is maintained at a high level.
[0189] For example, the output circuit **230** may count a number of rising edges of the oscillation signal OCS (or the sixth output signal OS**6**) that occur while the sixth operation signal CS**6** is maintained at a high level.
[0190] Furthermore, the output circuit **230** may determine a second threshold hold time HT**2** based on the number of rising edges of the oscillation signal OCS that are counted while the sixth operation signal CS**6** is maintained at a high level, and the period of the oscillation signal OCS.
[0191] For example, referring to FIG. **5**C, the output circuit **230** may determine that two rising edges of the oscillation signal OCS occur while the sixth operation signal CS**6** is maintained at a high level.
[0192] In addition, the output circuit **230** may determine that the second threshold hold time HT**2** is 200 ps based on the two rising edges of the oscillation signal OCS that occur while the sixth operation signal CS**6** is maintained at a high level, and the period of the oscillation signal OCS being 100 ps.
[0193] Referring to FIG. **5**B, the memory device **120** may output the output signal Dout having a high level (for example, “1”) in response to the input signal Din and the fifth delay clock DS**5**. It

will be understood that he input signal Din and the fifth delay clock DS**5** are in a state satisfying the hold time of the memory device **120**.

[0194] Referring to FIG. **5**C, the memory device **120** may output the output signal Dout having a low level (for example, "0") in response to the input signal Din and the sixth delay clock DS**6**. It will be understood that the input signal Din and the sixth delay clock DS**6** are in a hold time violation state that do not satisfy the hold time of the memory device **120**.

[0195] Accordingly, the test device **110**B may determine that the hold time of the memory device **120** has a value between the first threshold hold time HT**1** and the second threshold hold time HT**2**.

[0196] For example, the test device **110**B (or the control logic circuit **210**) may determine that the hold time of the memory device **120** has a value between a time difference between the rising edge of the fifth delay clock DS**5** and the falling edge of the input signal Din (corresponding to the first threshold hold time HT**1**) and a time difference between the rising edge of the sixth delay clock DS**6** and the falling edge of the input signal Din (corresponding to the second threshold hold time HT**2**).

[0197] Referring to the above-described configurations, the test device **110**B according to one or more example embodiments may determine the hold time of the memory device **120** based on a result of the XOR operation between the input signal Din and the delay clocks DS**5** and DS**6**, and a unit delay time.

[0198] According to one or more example embodiments, the test device **110**B (or the control logic circuit **210**) may apply different delay times to the input signal Din to generate a plurality of delayed input signals.

[0199] Furthermore, the test device **110**B may determine the hold time of the memory device **120** based on a result of an XOR operation between each of the plurality of delayed input signals and the clock signal CK, and the unit delay time.

[0200] As a result, the memory system **100**B according to one or more example embodiments may improve the accuracy of an operation (or process) of measuring the hold time of the memory device **120**.

[0201] FIG. **6**A is a circuit diagram illustrating a configuration of the first chain circuit according to one or more example embodiments. FIG. **6**B is a circuit diagram illustrating a configuration of a first internal multiplexer and the second internal multiplexer of the first chain circuit of FIG. **6**A.

[0202] Referring to FIG. **6**A, a first chain circuit **211**B according to one or more example embodiments may include delay elements C**1** to C**14**, each including at least one inverter.

[0203] According to one or more example embodiments, each of the delay elements C**1** to C**14** may include at least two inverters connected in series. For example, a first delay element C**1** may include two inverters connected in series. For example, a fourth delay element C**4** may include eight inverters connected in series.

[0204] In addition, the first chain circuit **211**B may include a plurality of internal multiplexers IM**1** to IM**20** electrically connected to the delay elements C**1** to C**14**.

[0205] For example, the first chain circuit **211**B may include a first internal multiplexer IM**1** electrically connected to the first delay element C**1** and the second delay element C**2**. In addition, the first chain circuit **211**B may include a second internal multiplexer IM**2** electrically connected to the first internal multiplexer IM**1**.

[0206] According to one or more example embodiments, the control logic circuit **210** may control the plurality of internal multiplexers IM**1** to IM**20**, included in the first chain circuit **211**B, to control a delay time applied to a clock signal CK.

[0207] For example, the control logic circuit **210** may control the plurality of internal multiplexers IM**1** to IM**20** such that the clock signal CK is delayed and output through at least a portion of the plurality of delay elements C**1** to C**14** included in the first chain circuit **211**B.

[0208] Referring to FIG. **6**B, the first internal multiplexer IM**1** may include a first input port IP**1**, a second input port IP**2**, and a first output port OP**1**.

[0209] The second internal multiplexer IM**2** may include a third input port IP**3**, a fourth input port IP**4**, and a second output port OP**2**. The second internal multiplexer IM**2** may receive a signal, output from the first output port OP**1**, through the third input port IP**3** and the fourth input port IP**4**.
[0210] According to one or more example embodiments, the first internal multiplexer IM**1** and the second internal multiplexer IM**2** may be implemented with substantially the same structure. Furthermore, the multiple internal multiplexers IM**1** to IM**20** may be implemented with substantially the same structure.
[0211] For example, an electrical distance between the third input port IP**3** and the second output port OP**2** may be the same as or similar to an electrical distance between the first input port IP**1** and the first output port OP**1**.
[0212] In addition, an electrical distance between the fourth input port IP**4** and the second output port OP**2** may be the same as or similar to an electrical distance between the second input port IP**2** and the first output port OP**1**.
[0213] In addition, the control logic circuit **210** according to one or more example embodiments may control the first internal multiplexer IM**1** and the second internal multiplexer IM**2** such that a signal, input to the first internal multiplexer IM**1**, is output through paths having different electrical distances within the first internal multiplexer IM**1** and the second internal multiplexer IM**2**.
[0214] For example, the control logic circuit **210** may control the first internal multiplexer IM**1** through a first signal S**1** to output a signal input to the first input port IP**1**, while controlling the second internal multiplexer IM**2** through a second signal S**2** to output the signal input to the fourth input port IP**4**.
[0215] For example, the control logic circuit **210** may control the second internal multiplexer IM**2** to output a signal input to the third input port IP**3** when controlling the first internal multiplexer IM**1** to output a signal input to the second input port IP**2**.
[0216] For example, the control logic circuit **210** may control electrically connected internal multiplexers (for example, IM**1** and IM**2**) to output signals input to input ports, which do not correspond to each other, at the internal multiplexers (for example, IM**1** and IM**2**).
[0217] In addition, the second chain circuit according to one or more example embodiments (for example, the second chain circuit **212** of FIG. **3**A) may have substantially the same structure as the first chain circuit **211**B.
[0218] Referring to the above-described configurations, the control logic circuit **210** (or the memory system **100**) according to one or more example embodiments may control electrically connected internal multiplexers such that a signal, output from each of the delay elements C**1** to C**14**, is output through paths having different electrical distances.
[0219] As a result, the memory system **100** according to one or more example embodiments may significantly reduce an error caused by mismatch between internal paths of each of the internal multiplexers IM**1** to IM**20** when controlling a delay of the clock signal CK through the chain circuits **211**B and **212**.
[0220] For example, the memory system **100** according to one or more example embodiments may control the delay of the clock signal CK through the above-described configurations to improve the accuracy of an operation (or process) of measuring timing specifications of the memory device **120**.
[0221] FIG. **7** is a circuit diagram illustrating a memory system including a test device further including a second multiplexer according to one or more example embodiments.
[0222] Referring to FIG. **7**, a memory system **100**C according to one or more example embodiments may include a test device **110**C and a memory device **120**. The test device **110**C may include a first chain circuit **211**, a second chain circuit **212**, a control logic circuit **210**, a first multiplexer MUX**1**, a second multiplexer MUX**2**, a first XOR gate XOR1, a second XOR gate XOR2, an oscillator **220**, an AND gate A, and an output circuit **230**.
[0223] The memory system **100**C illustrated in FIG. **7** may be understood as an example of the

memory system **100** illustrated in FIG. **1**. Therefore, the same or substantially the same components may be represented by the same reference numerals, and redundant descriptions are omitted to avoid repetition.

[0224] Referring to FIG. **7**, the test device **110**C according to one or more example embodiments may include the second multiplexer MUX**2** electrically connected to the first XOR gate XOR1 and the second XOR gate XOR2.

[0225] For example, the test device **110**C may include the second multiplexer MUX**2** electrically connected between the first XOR gate XOR1 and the second XOR gate XOR2, and the AND gate A.

[0226] The second multiplexer MUX**2** according to one or more example embodiments may selectively output a signal, output from the first XOR gate XOR1, and a signal, output from the second XOR gate XOR2, in response to a second select signal SEL**2**.

[0227] According to one or more example embodiments, the control logic circuit **210** may control the second multiplexer MUX**2** through the second select signal SEL**2** to selectively output the signal, output from the first XOR gate XOR1, and the signal, output from the second XOR gate XOR2, to the AND gate A through the second multiplexer MUX**2**.

[0228] For example, the control logic circuit **210** may transmit the signal, output from the first XOR gate XOR1, to the AND gate A through the second multiplexer MUX**2** such that the output circuit **230** determines an access time of the memory device **120** based on the signal output from the first XOR gate XOR1.

[0229] For example, the control logic circuit **210** may transmit the signal, output from the second XOR gate XOR2, to the AND gate A through the second multiplexer MUX**2** such that the output circuit **230** determines a setup time and/or a hold time of the memory device **120** based on the signal output from the second XOR gate XOR2.

[0230] Referring to the above-described configurations, the test device **110**C according to one or more example embodiments may measure the access time of the memory device **120**, the setup time, and/or the hold time of the memory device **120** based on the second select signal SEL**2**.

[0231] For example, the test device **110**C according to one or more example embodiments may determine the timing specifications (access time, setup time, and/or hold time) of the memory device **120** using a minimum circuit configuration.

[0232] Accordingly, the memory system **100**C may significantly reduce the configuration and/or area of a circuit for measuring the timing specifications of the memory device **120**.

[0233] FIG. **8** is a circuit diagram illustrating a memory system including a test device further including a third multiplexer according to one or more example embodiments.

[0234] Referring to FIG. **8**, a memory system **100**D according to one or more example embodiments may include a test device **110**D and a memory device **120**. In addition, the test device **110**D may include a first chain circuit **211**, a second chain circuit **212**, a control logic circuit **210**, a first multiplexer MUX**1**, a second multiplexer MUX**2**, a third multiplexer MUX**3**, a first XOR gate XOR1, a second XOR gate XOR2, an oscillator **220**, an AND gate A, and an output circuit **230**.

[0235] The memory system **100**D illustrated in FIG. **8** may be understood as an example of the memory system **100** illustrated in FIG. **1**. Therefore, the same or substantially the same components may be represented by the same reference numerals, and redundant descriptions are omitted to avoid repetition.

[0236] Referring to FIG. **8**, the test device **110**D according to one or more example embodiments may include the third multiplexer MUX**3** electrically connected between the AND gate A and the output circuit **230**.

[0237] For example, the third multiplexer MUX**3** may selectively output a first output signal OS**1** and a first inverted signal IOS**1** output from the AND gate A in response to a third select signal SEL**3**.

[0238] The first inverted signal IOS**1** may be understood as a signal obtained by inverting the first

output signal OS**1** through an inverter.
[0239] The control logic circuit **210** according to one or more example embodiments may control the third multiplexer MUX**3** through the third select signal SEL**3** to selectively output the first output signal OS**1** and the first inverted signal IOS**1** to the output circuit **230**.
[0240] For example, the control logic circuit **210** may control the third multiplexer MUX**3** through the third select signal SEL**3** to alternately output the first output signal OS**1** and the first inverted signal IOS**1** to the output circuit **230**.
[0241] Furthermore, the output circuit **230** may count a number of rising edges of a first average signal corresponding to an average of the first output signal OS**1**, output through the third multiplexer MUX**3**, and the first inverted signal IOS**1**.
[0242] Referring to the above-described configurations, the test device **110**D (or the output circuit **230**) according to one or more example embodiments may count the number of rising edges based on a signal corresponding to the average of the signal output from the AND gate A and the signal obtained by inverting the output signal.
[0243] Furthermore, the output circuit **230** may determine the timing specifications of the memory device **120** based on the number of the counted rising edges.
[0244] As a result, the memory system **100**D according to one or more example embodiments may significantly reduce the impact of errors, occurring in a time domain adjacent to an edge of the first output signal OS**1**, due to internal components of the test device **110**D.
[0245] FIG. **9** is a flowchart illustrating a test method for determining an access time of a memory device according to one or more example embodiments.
[0246] Referring to FIG. **9**, the test device **110** (or the memory system **100**) according to one or more example embodiments may determine an access time of the memory device **120** based on a result of an XOR operation between a clock signal CK and an output signal Dout.
[0247] The output signal Dout may be referred to as a data signal output from the memory device **120** in response to an input signal Din and the clock signal CK input to the memory device **120**.
[0248] In operation S**10**, the test device **110** according to one or more example embodiments may determine a period of an oscillation signal OCS.
[0249] For example, the test device **110** (or the output circuit **230**) may determine a period of the oscillation signal OCS by dividing the oscillation signal OCS output from the oscillator **220**.
[0250] In operation S**20**, the test device **110** according to one or more example embodiments may count the number of rising edges of the oscillation signal OCS that occur while the first operation signal CS**1** is maintained at a high level.
[0251] The test device **110** may output a first operation signal CS**1** as a result of an XOR operation of the clock signal CK and the output signal Dout.
[0252] In addition, the test device **110** may perform an AND operation on the first operation signal CS**1** and the oscillation signal OCS.
[0253] Furthermore, the test device **110** (or the output circuit **230**) may count the number of rising edges of the first output signal OS**1** output as a result of the AND operation of the first operation signal CS**1** and the oscillation signal OCS.
[0254] In operation S**30**, the test device **110** according to one or more example embodiments may determine the access time of the memory device **120**.
[0255] For example, the test device **110** may determine the access time of the memory device **120** based on the number of rising edges of the oscillation signal OCS, counted while the first operation signal CS**1** is maintained at a high level, and the period of the oscillation signal OCS.
[0256] For example, the test device **110** may determine the access time of the memory device **120** to be 500 ps based on five rising edges of the oscillation signal OCS that occur while the first operation signal CS**1** is maintained at a high level, and the period of the oscillation signal OCS being 100 ps.
[0257] The access time may be understood to correspond to the time taken for the memory device

120 to output the output signal Dout in response to the clock signal CK in a state in which the input signal Din is input to the memory device **120**.

[0258] For example, the output circuit **230** may determine that the time taken for the memory device **120** to output a rising edge of the output signal Dout in response to the rising edge of the clock signal, in the state in which the input signal Din is input to the memory device **120**, is 500 ps.

[0259] Referring to the above-described configurations, the test device **110** according to one or more example embodiments may determine the period of the oscillation signal OCS by dividing the oscillation signal OCS.

[0260] In addition, the test device **110** may determine the access time of the memory device **120** using the determined period and the result of the XOR operation between the clock signal CK and the output signal Dout.

[0261] As a result, the memory system **100** according to one or more example embodiments may improve the accuracy of the operation (or process) of measuring the access time of the memory device **120**.

[0262] FIG. **10** is a flowchart illustrating a method for determining a unit delay time of a first chain circuit and a second chain circuit included in a test device according to one or more example embodiments.

[0263] Referring to FIG. **10**, the test device **110** according to one or more example embodiments may determine a unit delay time caused by one of the plurality of inverters included in each of the first chain circuit **211** and the second chain circuit **212**.

[0264] In operation S**10**, the test device **110** according to one or more example embodiments may determine the period of the oscillation signal OCS.

[0265] For example, the test device **110** (or the output circuit **230**) may determine the period of the oscillation signal OCS by dividing the oscillation signal OCS output from the oscillator **220**.

[0266] In operation S**21**, the test device **110** according to one or more example embodiments may generate a first delayed clock DS**1** and a second delayed clock DS**2**.

[0267] For example, the test device **110** may generate a first delayed clock DS**1** and a second delayed clock DS**2**, in which different delay times are applied to the clock signal CK, by controlling the first chain circuit **211** and the second chain circuit **212**.

[0268] For example, the test device **110** may generate a first delayed clock DS**1**, to which a first delay time equal to a first integer multiple of the unit delay time is applied, by controlling the first chain circuit **211**. In addition, the test device **110** may generate a second delayed clock DS**2**, to which a second delay time equal to a second integer multiple of the unit delay time is applied, by controlling the second chain circuit **212**. The second integer may be smaller than the first integer, but example embodiments are not limited thereto.

[0269] In operation S**31**, the test device **110** according to one or more example embodiments may count the number of rising edges of the oscillation signal OCS that occur while the second operation signal CS**2** is maintained at a high level.

[0270] The test device **110** according to one or more example embodiments may output a second operation signal CS**2** as a result of an XOR operation on the first delayed clock DS**1** and the second delayed clock DS**2**.

[0271] In addition, the test device **110** may perform an AND operation on the second operation signal CS**2** and the oscillation signal OCS.

[0272] Furthermore, the test device **110** (or the output circuit **230**) may count the number of rising edges of the second output signal OS**2** output as a result of the AND operation on the second operation signal CS**2** and the oscillation signal OCS.

[0273] In addition, the test device **110** may determine that a time difference TD between the first delayed clock DS**1** and the second delayed clock DS**2** is 1200 ps, based on 12 rising edges of the oscillation signal OCS that occur while the second operation signal CS**2** is maintained at a high level, and the period of the oscillation signal OCS being 100 ps.

[0274] Furthermore, the test device **110** may determine a unit delay time based on a difference between the number of inverters, included in an electrical path through which the first delayed clock DS**1** is output, and the number of inverters included in an electrical path through which the second delayed clock DS**2** is output.
[0275] For example, the test device **110** may determine the unit delay time to be 200 ps by dividing 1200 ps, the time difference TD between the first and second delayed clocks DS**1** and DS**2**, by 6 when the two delayed clock DS**1** and DS**2** have a delay time difference corresponding to six inverters therebetween.
[0276] For example, the test device **110** may determine that the unit delay time caused by one of the plurality of inverters included in each of the first chain circuit **211** and the second chain circuit **212** is 200 ps.
[0277] Referring to the above-described configurations, the test device **110** according to one or more example embodiments may determine the unit delay time using the result of the XOR operation between delayed clocks DS**1** and DS**2**, to which different delay times are applied to the clock signal CK, and the period of the oscillation signal OCS.
[0278] As a result, the memory system **100** according to one or more example embodiments may accurately determine the unit delay time of a circuit, applying a delay to the clock signal CK, to determine timing specifications of the memory device **120**.
[0279] FIG. **11** is a flowchart illustrating a test method for determining a setup time of a memory device based on unit delay time according to one or more example embodiments.
[0280] Referring to FIGS. **10** and **11**, the test device **110** according to one or more example embodiments may determine the setup time of the memory device **120** based on the unit delay time and the result of the XOR operation between the delayed clock and the input signal.
[0281] In operation S**51**, the test device **110** according to one or more example embodiments may input an input signal Din, having a waveform transitioning from a low level to a high level, to the memory device **120**.
[0282] For example, the test device **110** may set a value of the output signal Dout, output from the memory device **120**, to "0" and then input the input signal Din, having the waveform transitioning from a low level to a high level, to the memory device **120**.
[0283] In operation S**61**, the test device **110** according to one or more example embodiments may reduce the delay time of the clock signal CK by an integer multiple of the unit delay time.
[0284] For example, the test device **110** may control the first chain circuit **211** to output a delayed clock, to which a delay time equal to an integer multiple of the unit delay time is applied, to the clock signal CK.
[0285] For example, the test device **110** may control the first chain circuit **211** to output a third delayed clock DS**3**, to which a third delay time equal to an integer multiple of the unit delay time is applied, to the clock signal CK. For example, the third integer may be referred to as "2," but example embodiments are not limited thereto.
[0286] In operation S**71**, the test device **110** according to one or more example embodiments may determine whether the output signal Dout is output with a high level.
[0287] For example, the test device **110** may determine whether the memory device **120** outputs the output signal Dout having a high level in response to the rising edges of the input signal Din and the third delayed clock DS**3**.
[0288] For example, when the memory device **120** outputs the output signal Dout having a high level (for example, "1") in response to the input signal Din and the third delayed clock DS**3**, the input signal Din and the third delayed clock DS**3** may be understood to be in a state satisfying the setup time of the memory device **120**.
[0289] Furthermore, the test device **110** may output a fourth delayed clock DS**4**, to which a fourth delay time equal to an integer multiple of the unit delay time is applied, to the clock signal CK based on the memory device **120** outputting the output signal Dout having a high level in response

to the rising edge of the third delayed clock DS**3**. For example, the fourth integer may be referred to as “4,” but example embodiments are not limited thereto.
[0290] In addition, for example, when the memory device **120** outputs the output signal Dout having a low level (for example, “0”) in response to the input signal Din and the fourth delayed clock DS**4**, the input signal Din and the fourth delayed clock DS**4** may be understood to be in a setup time violation state that does not satisfy the setup time of the memory device **120**.
[0291] In operation S**81**, the test device **110** may determine a first threshold setup time ST**1** by counting the number of rising edges of the oscillation signal OCS that occur while a third operation signal CS**3** is maintained at a high level. The third operation signal CS**3** may be referred to as a result of an XOR operation between the third delayed clock DS**3** and the input signal Din.
[0292] In operation S**91**, the test device **110** may determine a second threshold setup time ST**2** by counting the number of rising edges of the oscillation signal OCS that occur while a fourth operation signal CS**4** is maintained at a high level. The fourth operation signal CS**4** may be referred to as a result of an XOR operation between the fourth delayed clock DS**4** and the input signal Din.
[0293] According to one or more example embodiments, the setup time of the memory device **120** may have a value between the first threshold setup time ST**1** and the second threshold setup time ST**2**. For example, the test device **110** may determine that the setup time of the memory device **120** has a value between the first threshold setup time ST**1** and the second threshold setup time ST**2**.
[0294] Referring to the above-described configurations, the test device **110** may determine the setup time of the memory device **120** based on the result of the XOR operation between the input signal Din and the delayed clocks DS**3** and DS**4**, and the unit delay time.
[0295] As a result, the memory system **100** according to one or more example embodiments may improve the accuracy of the operation (or process) of measuring the setup time of the memory device **120**.
[0296] FIG. **12** is a flowchart illustrating a test method for determining a hold time of a memory device based on the unit delay time according to one or more example embodiments.
[0297] Referring to FIGS. **10** and **12**, the test device **110** according to one or more example embodiments may determine a hold time of the memory device **120** based on a unit delay time and a result of an XOR operation between a delayed clock and an input signal.
[0298] In operation S**52**, the test device **110** according to one or more example embodiments may input an input signal Din, having a waveform transitioning from a high level to a low level, to the memory device **120**.
[0299] For example, the test device **110** may set a value of the output signal Dout, output from the memory device **120**, to “0” and then input an input signal Din, having a waveform transitioning from a high level to a low level, to the memory device **120**.
[0300] In operation S**62**, the test device **110** according to one or more example embodiments may increase the delay time of the clock signal CK by an integer multiple of the unit delay time.
[0301] For example, the test device **110** may control the first chain circuit **211** to output a delayed clock, to which a delay time equal to an integer multiple of the unit delay time is applied, to the clock signal CK.
[0302] For example, the test device **110** may control the first chain circuit **211** to output a fifth delayed clock DS**5**, to which a fifth delay time equal to an integer multiple of the unit delay time is applied, to the clock signal CK.
[0303] In operation S**72**, the test device **110** according to one or more example embodiments may determine whether the output signal Dout is output with a high level.
[0304] For example, the test device **110** may determine whether the memory device **120** outputs the output signal Dout having a high level in response to the rising edge of the input signal Din and the fifth delayed clock DS**5**.
[0305] For example, when the memory device **120** outputs the output signal Dout having a high level (for example, “1”) in response to the input signal Din and a fifth delayed clock DS**5**, the input

signal Din and the fifth delayed clock DS**5** may be understood to be in a state satisfying the hold time of the memory device **120**.

[0306] Furthermore, the test device **110** may output a sixth delayed clock DS**6**, to which a sixth delay time equal to an integer multiple of the unit delay time is applied, to the clock signal CK based on the memory device **120** outputting the output signal Dout having a high level in response to the rising edge of the fifth delayed clock DS**5**.

[0307] In addition, for example, when the memory device **120** outputs the output signal Dout having a low level (for example, “0”) in response to the input signal Din and a sixth delayed clock DS**6**, the input signal Din and the sixth delayed clock DS**6** may be understood to be in a hold time violation state that does not satisfy the hold time of the memory device **120**.

[0308] In operation S**82**, the test device **110** may determine a first threshold hold time HT**1** by counting the number of rising edges of the oscillation signal OCS that occur while a fifth operation signal CS**5** is maintained at a high level. The fifth operation signal CS**5** may be referred to as a result of an XOR operation between the fifth delayed clock DS**5** and the input signal Din.

[0309] In operation S**92**, the test device **110** may determine a second threshold hold time HT**2** by counting the number of rising edges of the oscillation signal OCS that occur while a sixth operation signal CS**6** is maintained at a high level. The sixth operation signal CS**6** may be referred to as a result of an XOR operation between the sixth delayed clock DS**6** and the input signal Din.

[0310] According to one or more example embodiments, the hold time of the memory device **120** may have a value between the first threshold hold time HT**1** and the second threshold hold time HT**2**. For example, the test device **110** may determine that the hold time of the memory device **120** has a value between the first threshold hold time HT**1** and the second threshold hold time HT**2**.

[0311] Referring to the above-described configurations, the test device **110** may determine the hold time of the memory device **120** based on the result of the XOR operation between the input signal Din and the delayed clocks DS**5** and DS**6**, and the unit delay time.

[0312] As a result, the memory system **100** according to one or more example embodiments may improve the accuracy of an operation (or process) of measuring the hold time of the memory device **120**.

[0313] As described above, the test device **110** according to one or more example embodiments may measure the access time of the memory device **120** using the result of the XOR operation between the period of the oscillation signal OCS and the clock signal CK, and the output signal Dout.

[0314] As a result, the memory system **100** according to one or more example embodiments may improve the accuracy of the operation (or process) of measuring the access time of the memory device **120**.

[0315] In addition, the test device **110** according to one or more example embodiments may determine the unit delay time caused by each inverter included in each of the chain circuits **211** and **212** using the result of the XOR operation between the delayed clocks DS**1** and DS**2** to which different delay times are applied.

[0316] Furthermore, the test device **110** may determine the setup time of the memory device **120** based on the result of the XOR operation between the input signal Din and the delayed clocks DS**3** and DS**4**, and the unit delay time.

[0317] As a result, the memory system **100** according to one or more example embodiments may improve the accuracy of the operation (or process) of measuring the setup time of the memory device **120**.

[0318] In addition, the test device **110** according to one or more example embodiments may determine the hold time of the memory device **120** based on the result of the XOR operation between the input signal Din and the delayed clocks DS**5** and DS**6**, and the unit delay time.

[0319] As a result, the memory system **100** according to one or more example embodiments may improve the accuracy of the operation (or process) of measuring the hold time of the memory

device **120**.
[0320] As set forth above, a test device according to example embodiments may improve the accuracy of an operation of measuring timing specifications of a memory device.
[0321] While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. A test device for a memory device, the test device comprising: a first exclusive OR (XOR) gate configured to output a first operation signal by performing an XOR operation between a clock signal and an output signal output from the memory device; an oscillator configured to output an oscillation signal; and an output circuit configured to determine a period of the oscillation signal, wherein the output circuit is configured to: count a number of rising edges of the oscillation signal while the first operation signal is maintained at a high level; and determine an access time of the memory device based on the counted number of the rising edges, and a period of the oscillation signal.
2. The test device of claim 1, further comprising: a first chain circuit and a second chain circuit, each being configured to receive the clock signal and each comprising a plurality of inverters; and a control logic circuit connected to the first chain circuit and the second chain circuit, wherein the control logic circuit is configured to control the first chain circuit and the second chain circuit such that the first chain circuit and the second chain circuit output delayed clocks to which delay times equal to different integer multiples of a unit delay time are respectively applied, and wherein the unit delay time corresponds to a time delayed by one of the plurality of inverters.
3. The test device of claim 2, further comprising: a first multiplexer configured to output either one of a signal output from the second chain circuit and an input signal input to the memory device, based on a first select signal; and a second XOR gate configured to output a second operation signal by performing an XOR operation between a signal output from the first multiplexer and a signal output from the first chain circuit, wherein the control logic circuit is further configured to: control the first chain circuit to output a first delayed clock to which a first delay time equal to a first integer multiple of the unit delay time is applied; and control the second chain circuit and the first multiplexer to output a second delayed clock to which a second delay time equal to a second integer multiple of the unit delay time is applied, and wherein the output circuit is further configured to determine the unit delay time by counting a number of rising edges of the oscillation signal that occur while the second operation signal is maintained at a high level.
4. The test device of claim 3, wherein the control logic circuit, in a state in which the input signal transitions from a low level to a high level, is configured to: control the first chain circuit to output a third delayed clock to which a third delay time equal to a third integer multiple of the unit delay time is applied; based on the output signal having a high level being output from the memory device in response to a rising edge of the third delayed clock, control the first chain circuit to output a fourth delayed clock, to which a fourth delay time equal to a fourth integer multiple of the unit delay time is applied, wherein the output signal having a low level is output in response to the fourth delay time; and determine a setup time of the memory device based on the unit delay time, the third delay time, and the fourth delay time.
5. The test device of claim 4, wherein the output circuit is further configured to, in determining the setup time: determine a first threshold setup time by counting a number of rising edges of the oscillation signal that occur while a third operation signal, which is a result of an XOR operation between the third delayed clock and the input signal, is maintained at a high level; determine a second threshold setup time by counting a number of rising edges of the oscillation signal that occur while a fourth operation signal, which is a result of an XOR operation between the fourth

delayed clock and the input signal, is maintained at a high level; and determine the setup time to have a value between the first threshold setup time and the second threshold setup time.
6. The test device of claim 3, wherein the control logic circuit, in a state in which the input signal transitions from a high level to a low level, is configured to: control the first chain circuit to output a fifth delayed clock to which a fifth delay time equal to a fifth integer multiple of the unit delay time is applied; based on the output signal having a high level being output from the memory device in response to the fifth delayed clock, control the first chain circuit to output a sixth delayed clock, to which a sixth delay time equal to a sixth integer multiple of the unit delay time is applied, wherein the output signal having a low level is output in response to the sixth delayed clock; and determine a hold time of the memory device based on the unit delay time, the fifth integer multiple, and the sixth integer multiple.
7. The test device of claim 6, wherein the output circuit is configured to, in determining the hold time: determine a first threshold hold time by counting a number of rising edges of the oscillation signal that occur while a fifth operation signal, which is a result of an XOR operation between the fifth delayed clock and the input signal, is maintained at a high level; determine a second threshold hold time by counting a number of rising edges of the oscillation signal that occur while a sixth operation signal, which is a result of an XOR operation between the sixth delayed clock and the input signal, is maintained at a high level, and determine the hold time to have a value between the first threshold hold time and the second threshold hold time.
8. The test device of claim 3, further comprising: a second multiplexer electrically connected to the first XOR gate and the second XOR gate; and an AND gate electrically connected between the second multiplexer and the oscillator, and the output circuit, wherein the output circuit is further configured to count a number of rising edges output through the AND gate while a signal, output from the second multiplexer, of the oscillation signal is maintained at a high level.
9. The test device of claim 8, further comprising: a third multiplexer configured to receive a first output signal output from the AND gate and a first inverted version of the first output signal, wherein the output circuit is configured to count a number of rising edges of a first average signal corresponding to an average of the first output signal output from the third multiplexer and the first inverted version.
10. The test device of claim 2, wherein the first chain circuit comprises: a first internal multiplexer electrically connected to at least a portion of the plurality of inverters and comprising a first input port, a second input port, and a first output port; and a second internal multiplexer comprising a third input port and a fourth input port, through which a signal output from the first output port is received, and a second output port, wherein an electrical distance between the third input port and the second output port is the same as an electrical distance between the first input port and the first output port, wherein an electrical distance between the fourth input port and the second output port is the same as an electrical distance between the second input port and the first output port, and wherein the control logic circuit is configured to control the first internal multiplexer and the second internal multiplexer to output a signal, input to the first internal multiplexer, to the second output port through a path having different electrical distances in the first internal multiplexer and the second internal multiplexer.
11. A test method for a memory device, the test method comprising: determining a period of an oscillation signal, output from an oscillator; counting a number of rising edges of the oscillation signal that occur while a first operation signal, which is a result of an exclusive OR (XOR) operation between a clock signal and an output signal output from the memory device, is maintained at a high level; and determining an access time of the memory device based on the counted number of the rising edges, and the period of the oscillation signal.
12. The test method of claim 11, comprising: generating a first delayed clock, in which a first delay time equal to a first integer multiple of a unit delay time is applied to the clock signal, and a second delayed clock in which a second delay time equal to a second integer multiple of the unit delay

time is applied to the clock signal; counting a number of rising edges of the oscillation signal that occur while a second operation signal, which is a result of an XOR operation between the first delayed clock and the second delayed clock, is maintained at a high level; and determining the unit delay time based on the number of the rising edges of the oscillation signal counted while the second operation signal is maintained at the high level, and the period of the oscillation signal.

13. The test method of claim 12, further comprising: inputting an input signal, which transitions from a low level to a high level, to the memory device; inputting a third delayed clock, to which a third delay time equal to a third integer multiple of the unit delay time is applied, to the memory device; based on the output signal having a high level being output from the memory device in response to a rising edge of the third delayed clock, inputting a fourth delayed clock, to which a fourth delay time equal to a fourth integer multiple of the unit delay time is applied, to the memory device, wherein the output signal having a low level is output in response to the fourth delayed clock; and determining a setup time of the memory device based on the unit delay time, the third delay time, and the fourth delay time.

14. The test method of claim 13, wherein the determining the setup time comprises: determining a first threshold setup time by counting a number of rising edges of the oscillation signal that occur while a third operation signal, which is a result of an XOR operation between the third delayed clock and the input signal, is maintained at a high level; determining a second threshold setup time by counting a number of rising edges of the oscillation signal that occur while a fourth operation signal, which is a result of an XOR operation between the fourth delayed clock and the input signal, is maintained at a high level, determining the setup time to have a value between the first threshold setup time and the second threshold setup time.

15. The test method of claim 13, further comprising: inputting the input signal, which transitions from a high level to a low level, to the memory device; inputting a fifth delayed clock, to which a fifth delay time equal to a fifth integer multiple of the unit delay time is applied, to the memory device; based on the output signal having a high level being output from the memory device in response to the fifth delayed clock, inputting a sixth delayed clock, to which a sixth delay time equal to a sixth integer multiple of the unit delay time is applied, to the memory device, wherein the output signal having a low level is output in response to the sixth delayed clock; and determining a hold time of the memory device based on the unit delay time, the fifth delay time, and the sixth delay time.

16. A test device for a memory device, the test device comprising: an oscillator configured to output an oscillation signal; a first exclusive OR (XOR) gate configured to perform an XOR operation between a clock signal and an output signal output from the memory device; a first chain circuit and a second chain circuit configured to output delayed clocks in which different delay times are respectively applied to the clock signal; a second XOR gate configured to perform an XOR operation between a signal output from the first chain circuit and a signal output from the second chain circuit; and an output circuit configured to determine a period of the oscillation signal, wherein the output circuit is configured to: determine an access time of the memory device based on a number of rising edges of the oscillation signal that occur while a first operation signal output from the first XOR gate is maintained at a high level, and the period of the oscillation signal; and determine a unit delay time based on a number of rising edges of the oscillation signal that occur while a second operation signal output from the second XOR gate is maintained at a high level.

17. The test device of claim 16, further comprising: a control logic circuit connected to the first chain circuit and the second chain circuit, wherein the control logic circuit is configured to: control the first chain circuit to output a first delayed clock to which a first delay time, equal to a first integer multiple of the unit delay time, is applied; and control the second chain circuit to output a second delayed clock to which a second delay time, equal to a second integer multiple of the unit delay time, is applied, and wherein the second XOR gate is configured to output the second operation signal as a result of an XOR operation between the first delayed clock and the second delayed clock.

18. The test device of claim 17, wherein the control logic circuit, in a state in which the input signal transitions from a low level to a high level, is configured to: control the first chain circuit to output a third delayed clock to which a third delay time equal to a third integer multiple of the unit delay time is applied; based on the output signal having a high level being output in response to the third delayed clock, control the first chain circuit to output a fourth delayed clock, to which a fourth delay time equal to a fourth integer multiple of the unit delay time is applied, wherein the output signal having a low level is output in response to the fourth delayed clock; and determine a setup time of the memory device based on the unit delay time, the third integer multiple, and the fourth integer multiple.

19. The test device of claim 18, wherein the control logic circuit, in a state in which the input signal transitions from a high level to a low level, is configured to: control the first chain circuit to output a fifth delayed clock to which a fifth delay time equal to a fifth integer multiple of the unit delay time is applied; based on the output signal having a high level being output in response to the fifth delayed clock, control the first chain circuit to output a sixth delayed clock, to which a sixth delay time equal to a sixth integer multiple of the unit delay time is applied, wherein the output signal having a low level is output in response to the sixth delayed clock; and determine a hold time of the memory device based on the unit delay time, the fifth integer multiple, and the sixth integer multiple.

20. The test device of claim 16, further comprising: a second multiplexer electrically connected to the first XOR gate and the second XOR gate; and an AND gate electrically connected between the second multiplexer and the output circuit, wherein the output circuit is configured to count a number of rising edges output through the AND gate while a signal, output from the second multiplexer, of the oscillation signal is maintained at a high level.